Examiner only

**10.** (a) Some household cleaners are a concentrated solution of ammonia.

To determine the concentration of an ammonia solution, 10.0 $cm^3$ of a household cleaner was titrated with dilute sulfuric acid of concentration 1.5 mol / $dm^3$.

The end-point was determined by using the indicator methyl orange.

The procedure was repeated three times and the mean volume of dilute sulfuric acid needed to neutralise 10.0 $cm^3$ of the ammonia solution was found to be 12.0 $cm^3$.


(i) Use the equation below to calculate the number of moles of sulfuric acid in 12.0 $cm^3$ of the 1.5 mol / $dm^3$ solution. [2]

$$\text{concentration} = \frac{\text{number of moles}}{\text{volume}}$$

Number of moles of sulfuric acid = .................................................... mol

(ii) Ammonia solution reacts with sulfuric acid according to the equation below.

$$2NH_3(aq) + H_2SO_4(aq) \longrightarrow (NH_4)_2SO_4(aq)$$

Use the equation for the reaction to find the number of moles of ammonia in 10.0 $cm^3$ of the household cleaner. [1]

Number of moles of ammonia = .................................................... mol

(iii) Calculate the concentration of ammonia in mol / $dm^3$. [2]

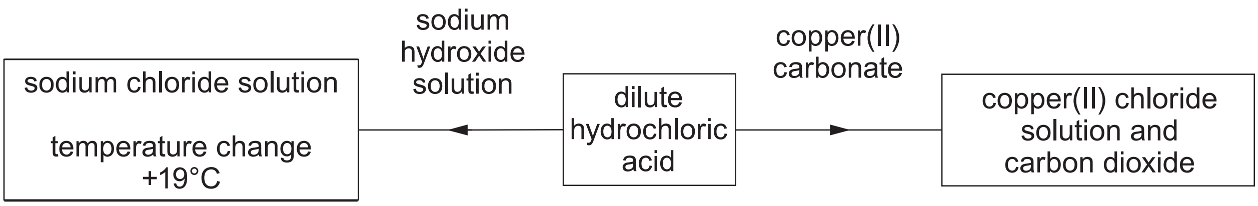
Concentration of ammonia = .................................................... mol / $dm^3$



 (3410UB0-1)

Examiner only

(b) The diagram shows two reactions of dilute hydrochloric acid.




(i) Predict the temperature change if dilute ethanoic acid, $CH_3COOH$, is added to sodium hydroxide instead of dilute hydrochloric acid. Give the reason for your answer. [2]

..............................................................................................................................................

..............................................................................................................................................

..............................................................................................................................................

(ii) A salt is formed when ethanoic acid is added to copper(II) carbonate.

I. Give the name of the salt.

.............................................................................................. [1]

II. The negative ion present in the salt is $CH_3COO^-$.

Underline the correct formula of the salt. [1]

**$Cu(CH_3COO)_2$** **$CuCH_3COO$** **$Cu_2CH_3COO$** **$Cu_2(CH_3COO)_2$**

9